Nose Wheel Fork – Maintenance Practices

1. Removal/Installation

Refer to Figure 1 for details.

Equipment Required:

- Locknut Tool Adapter T–187 (Heroux Machine Parts Ltd.) (for use with 3/4–inch square drive)

A. Remove Nose Wheel Fork

1) Jack aircraft until nose wheel is clear of ground. (Refer to 07-00-00, Lifting and Shoring – General Data).

2) Release air pressure from shock strut.

WARNING ENSURE ALL AIR PRESSURE IS RELEASED BEFORE REMOVING NOSE WHEEL FORK FROM NOSE GEAR SHOCK STRUT.

3) Remove nose wheel. (Refer to 32-40-61, Nose Wheel – General Data).

4) Remove nut, washer and bolt securing lower torque arm to fork and retain with torque arm for installation on replacement fork. On forks used with intermediate flotation tires retain the two additional Teflon washers.

5) If applicable, remove taxi light from fork.

6) Remove nut, washer and lockbolt.

7) Unscrew locknut, using special tool T–187, and remove locknut with washer and shim. Remove diaphragm and remove fork from piston tube.

B. Install Nose Wheel Fork

1) Position piston tube in fork, aligning slot with holes in bushings. Ensure that cartridge assembly is installed within piston tube and insert diaphragm. Lubricate surface of fork to be contacted by locknut with machine oil and install shim, washer and locknut to piston tube.

Note To maintain alignment between piston tube and bushings during following locknut tightening procedure, it is recommended that a headed pin, 0.25 inch diameter and approximately 0.625 inch in length be inserted through bushings in fork and slot in piston tube.

2) Using special tool T–187, torque locknut to 400 pound–feet. If the elongated hole in the locknut is visible through the bushing, increase torque until holes in locknut and bushing align.

Note If hole in locknut is not visible through bushing after initial torque loading, reduce thickness of shim as necessary and repeat tightening procedure.

3) Remove alignment pin (refer to Para 1.B.1)) if used.

4) Coat bores of bushings, lockbolt and washer with Chromate Epoxy Primer to MIL–PRF–23377 and install in fork while primer is still wet. Lock nut with cotter pin.

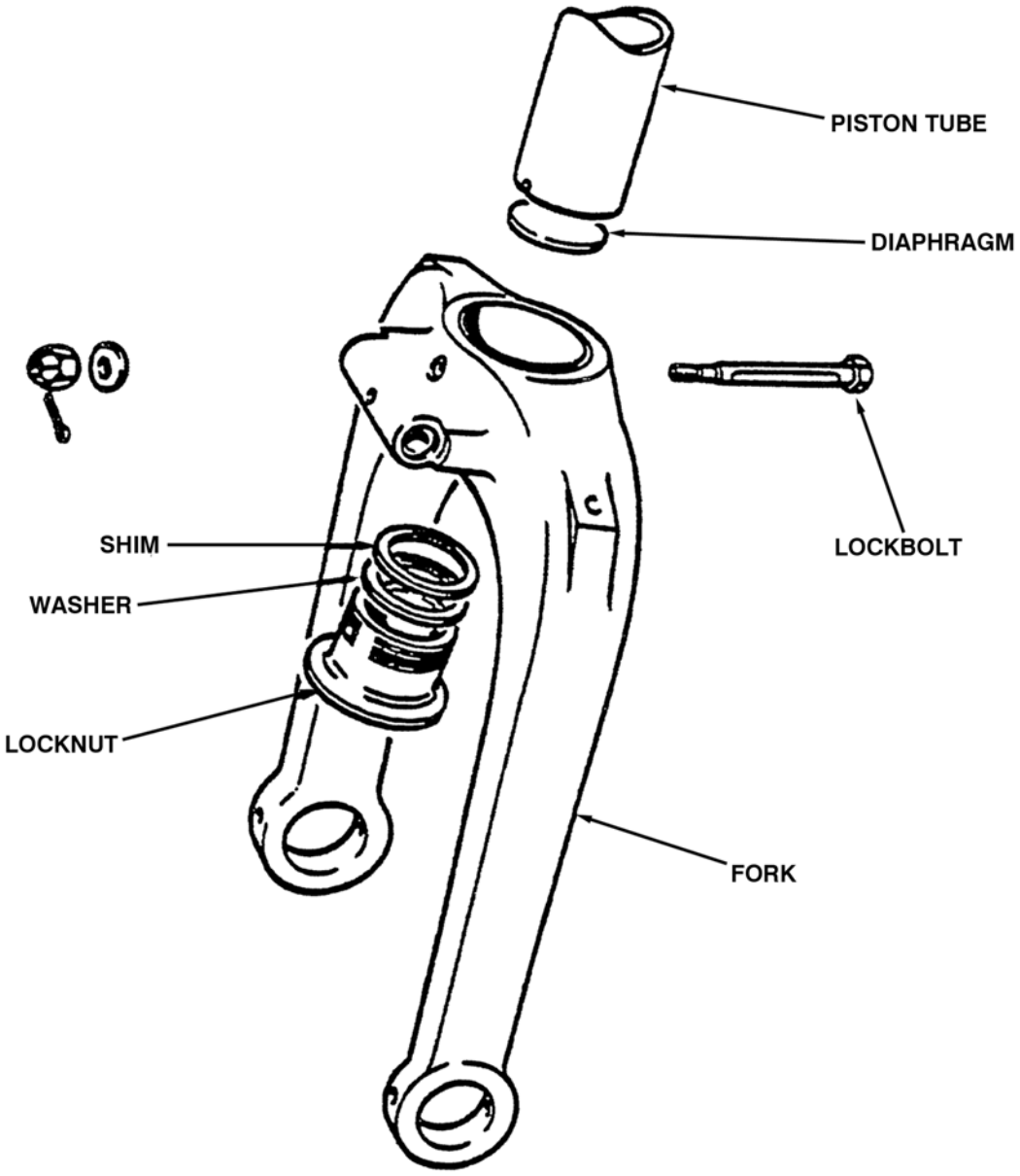
5) If applicable, install taxi light on fork.

6) Attach lower torque arm to fork with bolt, washer and nut. On fork used with intermediate flotation tires, install two Teflon washers, one at each lower torque arm lug, between torque arm and fork. Finger tighten nut, tighten to next slot and install cotter pin. Check torque arm for freedom of movement.

7) Service shock strut. (Refer to 12-10-20, Landing Gear – General Data).

8) Install nose wheel. (Refer to 32-40-61, Nose Wheel – General Data).

9) Lower aircraft to ground.



Nose Wheel Fork – Removal and Installation
Figure 1